**DATA flow diagram**

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|---|---|
| **Date** | 30 September 2026 |
| Team id | SWTID-2026-2551 |
| Project name | Fit Buddy – AI Fitness Plan Generator using Gemini Models |
| Maximum mark | 2marks |

**External Entity**

**User**

The user provides personal information such as name, age, weight, fitness goal, and workout intensity.

**process 1 – Validate User Input**

Receives and validates the user's fitness information.

**Process 2 – Generate Fitness Plan**

Sends the validated user information to the Gemini AI model and generates a personalized 7-day workout plan.

**Process 3 – Generate Nutrition Tip**

Generates a suitable nutrition or recovery tip based on the user's fitness goal

**Process 4 – Update Plan**

**Data Store**

Receives user feedback and updates the existing workout plan using Gemini AI.

**SQLite Database**

Stores user details, original workout plans, and updated workout plans.

**Data Flow**

**User → Validate User Input**

Name, Age, Weight, Goal, Intensity

**Validate User Input → Generate Fitness Plan**

Validated User Information

**Generate Fitness Plan → Gemini AI**

Fitness Requirements

**Gemini AI → Generate Fitness Plan**

Personalized 7-Day Workout Plan

**Generate Fitness Plan → SQLite Database**

User Details + Workout Plan

**Generate Fitness Plan → User**

Personalized Workout Plan

**User → Update Plan**

Feedback

**Update Plan → Gemini AI**

Original Plan + User Feedback

**Gemini AI → Update Plan**

Updated Workout Plan

**Update Plan → SQLite Database**

Updated Plan

**Update Plan → User**

Updated Workout Plan

This follows the uploaded template's DFD concept: data moves between **external entities, processes, data stores, and labeled data flows**.